

16 PBRs

Why are we learning this?

In the previous lecture, we studied Plug Flow Reactors (PFRs). Real catalytic reactors are often not empty tubes. Instead, they contain solid catalyst particles packed inside the reactor. These systems are called **Packed Bed Reactors (PBRs)**.

Packed beds introduce a new phenomenon: Pressure Drop

As fluid moves through the packed catalyst particles, friction causes pressure to decrease. This pressure change affects concentration, reaction rate, and reactor performance.

By the end of today's lesson, you should understand:

- How PBRs differ from PFRs.
- Why pressure drop occurs.
- How pressure drop influences reactor performance.
- The purpose of the Ergun equation.
- Why PBR design often leads to coupled differential equations.
- Why numerical methods are frequently needed for PBR design.

Course Roadmap

Introduction to CRE



General Balance Equation



Common Reactor Models
(CSTR, Batch, PFR)



Reaction Progress
(Conversion)



Graphical Reactor Design

(Inverse Rate/Levenspiel Plots)



Rate Laws

(Kinetics)



Equilibrium

(Reversible Reactions)



Equilibrium Conversion

(Thermodynamic Limitations)



One Reaction: Solving in terms of Reaction Progress



► One Reaction: Reactor Design ◀

(CSTR, Batch, PFR, PBR)



Catalytic Reactions



Numerical Methods

(Python)



Energy Balances



One Reaction: Non-Isothermal Reactor Design



Multiple Reactions



Reactor Safety



Challenging the Well-Mixed Assumption

(Residence Time Distributions)



Challenging the Elementary Rate Law Assumption

(Reaction Mechanisms)

Big Idea

PBR $\approx$ PFR + Pressure Drop

Much of today's lecture is devoted to understanding the consequences of that extra term.

PBR mole balance:

- Let's review PFRs by doing an example: Gas phase reaction $A + 0.5 B \rightarrow C$, carried out in an isothermal, isobaric PFR. Determine the V needed for 60% conversion. The feed is stoichiometric in A & B. Additionally:

- $-r_A = k P_A^{\frac{1}{3}} P_B^{\frac{2}{3}}$, $k = 0.00392 \text{ mol/L} \cdot \text{atm} \cdot \text{s}$, $F_{A0} = 0.1362 \text{ mol/s}$, $P_0 = 10 \text{ atm}$

PFR mole balance:

Rate law:

Expressions for P_i :

δ and ϵ :

Limits of integration:

Solution: $V = 6.71$ L. You should verify that you can get this solution. Try it!

Why Start with a PFR?

A packed bed reactor behaves very similarly to a plug flow reactor.

Both systems:

- Have continuous flow.
- Experience changing conditions along their length.
- Require differential reactor balances.

The major new feature of a PBR is pressure drop.

Understanding the PFR solution first makes it easier to see what pressure drop changes.

PFRs and PBRs Are Closely Related

Similarities

Both reactors:

- Operate continuously.

- Experience changing concentration with position.
- Use differential reactor balances.

Differences

PBRs contain solid catalyst particles.

As fluid flows through the packed catalyst:

- Friction occurs.
- Pressure decreases.
- Concentrations are affected.

These effects must be included in the reactor model.

What Is a Packed Bed Reactor?

A packed bed reactor contains many solid catalyst particles packed into a reactor vessel. In this course, it is reasonable to visualize the particles as small spheres. Fluid must weave around these particles as it flows through the reactor. Those interactions are responsible for pressure losses.

- What is meant by "packed bed"?

- Note: In this class, it's OK to think about packing as spherical pellets.
- Flow around the particle causes
- Drag causes
- Decreased pressure causes
- Decreased concentration influences

The Pressure-Drop Chain of Events

Pressure drop affects reactor performance through a sequence of connected effects:

Flow around particles



Friction / Drag



Pressure Drop



Lower Gas Density



Lower Concentration



Lower Reaction Rate

This chain of reasoning is the most important physical concept in today's lecture.

Why Is Pressure Drop Important?

For ideal gases:

$$C = P / RT$$

If pressure decreases: Concentration decreases.

If concentration decreases: Reaction rate often decreases.

Pressure drop therefore affects reactor performance even when temperature is constant.

- Account for pressure drop using

- Ergun Equation (full expression): $\frac{dP}{dz} = -\frac{G}{\rho g_c D_p} \left(\frac{1-\phi}{\phi^2} \right) \left[\frac{150(1-\phi)\mu}{D_p} + 1.75G \right]$
 - In this equation:
 - Important terms:

Why Do We Need the Ergun Equation?

The mole balance predicts how composition changes. The Ergun equation predicts how pressure changes.

Together they describe: Reaction Progress + Pressure Drop

Both effects occur simultaneously in a packed bed reactor.

- Ergun Equation, simplified forms that are useful for CRE/CRD (Equations 5-28 and 5-30 in Fogler):

- Hence, relationship between D_p and α :
- Application:
- Solving this set of coupled ODEs usually requires a numerical solution using software. EXCEPTION: $\epsilon = 0$ and $\frac{T}{T_0} = 1$. This can be solved analytically.

Why Are PBR Calculations More Difficult?

In a PFR with constant pressure, one ODE may be sufficient.

In a packed bed reactor, conversion changes AND pressure changes. These quantities influence one another. As a result, we often obtain coupled ODEs that must be solved simultaneously.

This Is Why We Need Python

Many packed bed reactor models do not have analytical solutions. Instead, engineers use numerical methods to solve the coupled differential equations. This is one reason numerical tools become increasingly important as reactor models become more realistic.

- Let's repeat the example from the beginning of the lecture, this time it's a PBR with pressure drop, i.e., it's not isobaric. Other changes:
- $k = 0.00392 \text{ mol/atm*kgcat*s}$, $\alpha = 0.0367 \text{ kg}^{-1}$

Mole balance:

What else is different?

What Changed Between the Two Problems?

The original problem assumed: Isothermal, Isobaric PFR

The new problem assumes: Isothermal, PBR with Pressure Drop

The chemistry is essentially the same. The difference is that pressure is now changing throughout the reactor. This allows us to isolate the effect of pressure drop on reactor performance.

Summary and Key Takeaways

Today we introduced Packed Bed Reactors (PBRs).

Key Ideas

- ✓ Packed bed reactors are closely related to PFRs.
- ✓ Catalyst particles create friction that causes pressure drop.
- ✓ Pressure drop decreases gas concentration and can reduce reaction rates.
- ✓ The Ergun equation is used to model pressure drop in packed beds.
- ✓ PBR design often requires solving coupled differential equations.
- ✓ Numerical methods are frequently required for realistic packed bed reactor calculations.

Most Important Idea

Packed Bed Reactor = Plug Flow Reactor + Pressure Drop

Understanding how pressure influences concentration and reaction rate is the key to understanding packed bed reactor behavior.